A Proposed Proof of the Binary-Form Moat Classification

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Abstract

We present a proof of the proposed classification of homogeneous binary integer forms whose nonzero values with a bounded number of distinct prime factors have uniformly bounded components at every fixed step size. The exceptional forms are precisely the nonzero integer multiples of powers of one linear form. The argument proves a finite-sieve obstruction for irreducible forms of arbitrary degree, using random root congruences, geometric entropy enrichment, coverage transfer, and a common sampling schedule. Disjoint prime blocks then handle every finite prime-factor budget. All estimates are uniform in the walk and its starting point.

1 The conjecture and the finite-sieve theorem

For $n \geq 1$, let $\omega(n)$ count the distinct rational prime divisors of n, so $\omega(1) = 0$. Multiplicities are not counted. For $F \in \mathbb{Z}[X, Y]$ and an integer $r \geq 0$, put

$$U_F(r) = \{z \in \mathbb{Z}^2 : F(z) \neq 0,\ \omega(|F(z)|) \leq r\}.$$

Join distinct points of this set when their Euclidean distance is at most D. We write $\mathrm{UM}(F)$ when, for every finite $D \geq 0$ and every $r \geq 0$, the sizes of all connected components have a finite common bound $B(F, D, r)$. The bound must be independent of the initial vertex.

Conjecture 1.1 (Binary-form moat classification)**.** Let $F \in \mathbb{Z}[X, Y]$ be nonzero and homogeneous of positive degree. Then $\mathrm{UM}(F)$ holds if and only if F is not of the form

$$F(X, Y) = c(aX + bY)^m, \qquad c \neq 0, \quad (a, b) \neq (0, 0), \quad m \geq 1,$$

with $c, a, b \in \mathbb{Z}$ and $m \in \mathbb{N}$.

The proof below establishes the classification by proving its higher-degree finite-sieve step within the argument. Throughout, "irreducible integer polynomial" means irreducible in $\mathbb{Z}[T]$. For $f(T) = \sum_{j=0}^{d} a_j T^j$ of degree d, write

$$f^{\mathrm{h}}(X, Y) = \sum_{j=0}^{d} a_j X^j Y^{d-j}.$$

For $H = f^{\mathrm{h}}$ and a finite set P of rational primes, define

$$\mathcal{A}_H(P) = \{z \in \mathbb{Z}^2 : p \nmid H(z) \text{ for every } p \in P\}.$$

It is periodic under $Q\mathbb{Z}^2$, where $Q = \prod_{p \in P} p$.

Theorem 1.2 (Generalized finite sieve). *Let $f \in \mathbb{Z}[T]$ be irreducible of degree $d \geq 2$ and put $H = f^{\mathrm{h}}$. For every $D \geq 1$ and every finite set S of rational primes there is a finite set P, disjoint from S, such that $\mathcal{A}_H(P)$ contains no infinite self-avoiding walk with steps at most D. The set P is chosen independently of the walk.*

Theorem 1.2 is proved in Sections 3–8. The argument fixes an arbitrary deterministic walk; only its sampled times and auxiliary residue maps are random. The finite-sieve theorem is an intermediate conclusion of this proof, not an additional conjectural assumption. For $D < 1$, all lattice components are singletons.

2 Elementary reductions and finite budgets

Lemma 2.1 (Divisors, content, and powers). *If $G \mid F$ and $\mathrm{UM}(G)$ holds, then $\mathrm{UM}(F)$ holds. For $c \in \mathbb{Z} \setminus \{0\}$ and $m \geq 1$,*

$$\mathrm{UM}(cF) \iff \mathrm{UM}(F), \qquad \mathrm{UM}(F^m) \iff \mathrm{UM}(F).$$

Proof. If $F(z) \neq 0$ and $G \mid F$, then $G(z) \neq 0$ and every prime dividing $G(z)$ divides $F(z)$. Thus $U_F(r) \subseteq U_G(r)$; each path in the smaller set remains a path in the larger set. Also

$$U_{cF}(r) \subseteq U_F(r) \subseteq U_{cF}(r + \omega(|c|)).$$

The second inclusion uses $\omega(|uv|) \leq \omega(|u|) + \omega(|v|)$ for nonzero integers. Quantifying over all budgets proves the assertion for content. Finally $U_{F^m}(r) = U_F(r)$ because positive powers have the same prime divisors and the same zero set. □

Proposition 2.2 (The exceptional obstruction). *If $F = c(aX + bY)^m$ as in Conjecture 1.1, then $\mathrm{UM}(F)$ fails.*

Proof. The distinct points $z_n = (a+bn, b-an)$, $n \geq 0$, have constant nonzero value $q = c(a^2+b^2)^m$. Consecutive points have distance $\sqrt{a^2+b^2}$. Taking $r = \omega(|q|)$ gives an infinite path in one component of $U_F(r)$. □

Lemma 2.3 (Periodic forbidden intervals). *For integers $r \geq 0$ and $W \geq 1$ there are $A \in \mathbb{Z}$ and $Q \geq 1$ such that any nonzero $n \in \mathbb{Z}$ with $\omega(|n|) \leq r$ has $(n - A) \bmod Q \geq W$, with remainder in $\{0, \ldots, Q-1\}$.*

Proof. Choose distinct primes $p_{i,j}$ for $0 \leq i < W$, $0 \leq j \leq r$, and let $Q = \prod_{i,j} p_{i,j}$. By the Chinese remainder theorem choose $A \equiv -i \pmod{p_{i,j}}$. If $n = A + kQ + i$ with $0 \leq i < W$, then all $r+1$ primes $p_{i,0}, \ldots, p_{i,r}$ divide n. This contradicts the budget when $n \neq 0$. □

Theorem 2.4 (Two independent linear divisors)**.** *Suppose $aX + bY$ and $cX + eY$ divide $F \in \mathbb{Z}[X, Y]$, and $ae - bc \neq 0$. Then* UM(F) *holds.*

Proof. Fix D, r and choose an integer $W \geq \max\{1, D(|a| + |b|), D(|c| + |e|)\}$. Both linear coordinates change by at most W on a D-edge. At an allowed point they are nonzero and have prime-factor budget at most r. With A, Q from Lemma 2.3, either coordinate lies in

$$[A + kQ + W, A + (k + 1)Q - 1] \cap \mathbb{Z} \quad \text{for some } k \in \mathbb{Z}.$$

Permitted integers in distinct such intervals are at least $W + 1$ apart. Thus both interval indices remain constant on a component. The two linear coordinates jointly define an injective map on $\mathbb{Z}^2$ because the determinant is nonzero. Each fixed pair of intervals has at most Q^2 integer coordinate pairs. This bounds every component. □

Proposition 2.5 (Algebraic dichotomy)**.** *Every nonexceptional nonzero homogeneous form F of positive degree has either two independent integer linear divisors or a divisor q^{h} with $q \in \mathbb{Z}[T]$ irreducible and* $\deg q \geq 2$.

Proof. Put $m = \deg F$, $f(T) = F(T, 1)$, and $n = \deg f$. Then $f \neq 0$, $n \leq m$, and $F = Y^{m-n} f^{\mathrm{h}}$. Factor

$$f = c \prod_{i=1}^{s} q_i^{e_i}, \qquad F = cY^{m-n} \prod_{i=1}^{s} (q_i^{\mathrm{h}})^{e_i},$$

where $c \neq 0$ is integral and the q_i are nonconstant pairwise nonassociate irreducibles in $\mathbb{Z}[T]$. Homogenization commutes with products at the sum of their degrees, proving the second identity. If a q_i has degree at least two, it supplies the asserted divisor. Otherwise all q_i are primitive linear polynomials. Two distinct ones cannot be proportional over $\mathbb{Q}$, since proportional primitive integer polynomials differ only by a sign. Thus $s \geq 2$ gives two independent linear divisors. If $s = 1$ and $m > n$, its homogenization and Y are independent. The cases left are $F = cY^m$ and $F = c(aX + bY)^m$, precisely the exceptions. □

Lemma 2.6 (Periodicity gives a uniform bound)**.** *Suppose $A \subseteq \mathbb{Z}^2$ is invariant under $Q\mathbb{Z}^2$, $Q \geq 1$, and has no infinite self-avoiding D-step walk. Its D-components have at most Q^2 vertices and diameter at most $(Q^2 - 1)D$.*

Proof. The graph is locally finite. An infinite component would give a finitely branching tree of finite simple paths from one vertex, with arbitrarily deep nodes. Successively choosing a child with arbitrarily deep descendants produces an infinite simple path. Thus all components are finite. If distinct x, y in a component C have equal residues modulo Q, translation by $v = y - x \in Q\mathbb{Z}^2$ sends C to the component containing y, so $C + v = C$. Iteration contradicts finiteness. Hence C injects into $(\mathbb{Z}/Q\mathbb{Z})^2$. A shortest path has at most $|C| - 1$ edges, giving the diameter bound. □

Lemma 2.7 (A finite union step)**.** *If the D-components of A have diameter at most a, and the $(a+2D)$-components of B have diameter at most b, then the D-components of $A \cup B$ have diameter at most $b + 2a + 2D$.*

Proof. On a finite D-path in the union, consecutive visits to B are either adjacent, or separated by a segment in a single D-component of A. They are therefore at distance at most $a + 2D$. All visits lie in one $(a + 2D)$-component of B. Each endpoint of the path is within $a + D$ of the first or last visit. The triangle inequality gives the bound. A path with no visit to B lies in one component of A. □

Proposition 2.8 (From the finite sieve to every finite budget)**.** *Theorem 1.2 implies* $\mathrm{UM}(f^{\mathrm{h}})$ *for every irreducible $f \in \mathbb{Z}[T]$ of degree at least two.*

Proof. Fix $D \geq 1, r \geq 0$, and write $H = f^{\mathrm{h}}$. Apply the finite-sieve theorem to obtain P_1 at step bound D. Periodicity gives a diameter bound a_1 for $\mathcal{A}_H(P_1)$. Suppose $P_1, \ldots, P_j$ are disjoint and their avoiding-set union has D-component diameter at most a_j. Apply the theorem at step bound $a_j + 2D$, excluding $P_1 \cup \cdots \cup P_j$, to obtain P_{j+1}. Periodicity bounds its components at this enlarged step bound by some diameter b_{j+1}. Lemma 2.7 bounds the new union at step D by $a_{j+1} = b_{j+1} + 2a_j + 2D$. Continue until there are $r + 1$ disjoint blocks. Every point of $U_H(r)$ avoids at least one whole block: otherwise the $r + 1$ disjoint blocks supply distinct prime divisors of $H(z) \neq 0$. Thus

$$U_H(r) \subseteq \bigcup_{j=1}^{r+1} \mathcal{A}_H(P_j).$$

This union has bounded-diameter components, hence finite components, and is periodic modulo $Q = \prod_{p \in \cup_j P_j} p$. Lemma 2.6 gives the uniform bound Q^2, also valid for the smaller graph on $U_H(r)$. □

3 Arithmetic, walks, and entropy

We record the conventions needed to sample a fixed walk and to compare the resulting residue distributions. All logarithms are natural. The notation $A \ll_{f,D} B$ means $A \leq c_{f,D} B$ with a constant depending only on f, D; $A \asymp_{f,D} B$ means both inequalities. In the finite-sieve argument f is fixed; constants may depend on f and D. Asymptotic statements about a collection of walks or starting times will always be uniform over that collection.

3.1 Irreducible forms and splitting primes

For the remainder of the finite-sieve proof fix an irreducible $f(T) = \sum_{j=0}^{d} a_j T^j \in \mathbb{Z}[T]$ with $d \geq 2$, put $H = f^{\mathrm{h}}$, and set $C_f = 1 + \sum_j |a_j|$. The polynomial f is primitive and irreducible over $\mathbb{Q}$. For every nonzero $v = (x, y) \in \mathbb{Z}^2$,

$$0 < |H(v)| \leq C_f \|v\|_\infty^d \leq C_f |v|^d. \tag{3.1}$$

Indeed, when $y \neq 0$ a zero would give the rational root x/y of f; when $y = 0$ the value is $a_d x^d \neq 0$. The upper bound follows by expanding the homogeneous polynomial and taking absolute values.

Let $L/\mathbb{Q}$ be the splitting field. Outside a finite set of primes, including those dividing the leading coefficient and the discriminant, complete splitting in L implies that the reduction of f has exactly d distinct roots in $\mathbb{F}_p$. This follows by reducing its distinct roots at a prime of L: their denominators and pairwise differences exclude only finitely many rational primes, and complete splitting makes the residue field $\mathbb{F}_p$. The Chebotarev prime-counting theorem gives density $\kappa_f = 1/[L:\mathbb{Q}] > 0$ for these primes, in the form $\pi_{\text{split}}(x) \sim \kappa_f x/\log x$. Fix also the finite excluded set S in Theorem 1.2. Let $\mathcal{B}_T$ consist of the completely splitting good primes in $[T, 2T]$ outside S. Then

$$k_T := |\mathcal{B}_T| = (\kappa_f + o(1))\frac{T}{\log T}, \qquad L_T^* := \frac{1}{k_T}\sum_{p\in\mathcal{B}_T} \log p \in [\log T, \log(2T)]. \tag{3.2}$$

Subtract the counting asymptotic at T and $2T$ to obtain the first formula; finite exclusions and endpoints do not affect it.

Enumerate the roots as $\rho_{p,1}, \dots, \rho_{p,d}$ and define

$$\phi_{p,j}(x, y) = x - \rho_{p,j} y \in \mathbb{F}_p.$$

A root coordinate π means a pair (p, j); we use $z \bmod \pi$ as notation for $\phi_{p,j}(z)$. Each map is additive and onto $\mathbb{F}_p$. Factorization modulo p gives

$$H(x, y) = a_d \prod_{j=1}^{d} (x - \rho_{p,j} y) \pmod{p}. \tag{3.3}$$

Thus $p \nmid H(z)$ precisely when all d root coordinates are nonzero. Root labels at distinct primes will always be chosen independently and uniformly from $\{1, \dots, d\}$.

Lemma 3.1 (Root arithmetic)**.** *For a finite good-prime set P and primitive $v \in \mathbb{Z}^2$, put $P_\sigma(v) = \prod_{p\in P:\phi_{p,\sigma_p}(v)=0} p$. Then*

$$\mathbb{E}_\sigma \log P_\sigma(v) \le \frac{1}{d}\log|H(v)| \le \log|v| + \frac{\log C_f}{d}.$$

If $w \neq 0$ and $\phi_{p,j}(w) = 0$, then $|w| \ge (p/C_f)^{1/d}$. If two vectors vanish in the same root coordinates at a set P_0 of distinct primes, then $\prod_{p\in P_0} p$ divides their determinant.

Proof. Primitivity means the coordinates of v cannot both vanish modulo p. The equation $x = \rho y$ therefore holds for at most one root, so its probability is at most $1/d$. A hit implies $p \mid H(v)$ by (3.3). The product of these eligible distinct primes divides the nonzero integer $H(v)$, proving the expectation bound. The growth bound (3.1) proves the second inequality and, from $p \mid H(w) \neq 0$, the asserted kernel length. For the last assertion, the two vectors reduce to multiples of $(\rho, 1)$ at each common coordinate, so their determinant vanishes modulo that prime. Distinct primes are coprime, so their product divides it. □

3.2 Entropy conventions

For a finite variable X with law p, $H(X) = -\sum_x p(x)\log p(x)$, with $0\log 0 = 0$. We use conditional entropy and conditional mutual information in their usual finite-alphabet forms:

$$I(X; Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z).$$

We use the chain rule, subadditivity, and the fact that conditioning decreases entropy. These follow directly from the finite definitions and the log-sum inequality. When residue maps are selected randomly, we first fix those maps and compute entropy in the walk-sampling experiment; only afterward do we average over the maps. An expectation outside an entropy in this usage is an average of entropies, not the entropy of a mixture that includes the map choices.

For laws p, q on a finite set, put $\mathrm{TV}(p,q) = \frac{1}{2}\sum_x |p(x)-q(x)|$. Relative entropy is $D_{\mathrm{KL}}(p\|q) = \sum_x p(x)\log(p(x)/q(x))$, with the usual value $+\infty$ if q vanishes where $p > 0$. In particular, on an alphabet of size a,

$$D_{\mathrm{KL}}(p\|\mathrm{Unif}) = \log a - H(p), \qquad D_{\mathrm{KL}}(p\|q) \geq 2\,\mathrm{TV}(p,q)^2.$$

The second inequality is Pinsker's inequality in natural logarithms.

For completeness, let P, Q be finite probability laws and put $B = \{x : P(x) > Q(x)\}$, $u = P(B)$, and $v = Q(B)$. The log-sum inequality, from convexity of $t\log t$, gives

$$D_{\mathrm{KL}}(P\|Q) \geq D_{\mathrm{KL}}(\mathrm{Ber}(u)\|\,\mathrm{Ber}(v)).$$

Here $\mathrm{Ber}(u)$ is the Bernoulli law with success probability u. As a function of u, the expression on the right has value and first derivative zero at $u = v$, and second derivative $1/[u(1-u)] \geq 4$. It is therefore at least $2(u-v)^2 = 2\,\mathrm{TV}(P,Q)^2$, with boundary cases by continuity.

We shall also use $h(u) = -u\log u - (1-u)\log(1-u)$ for binary entropy, with its continuous values at 0 and 1.

3.3 Sampling a deterministic walk

Fix $D \geq 1$ and an infinite deterministic sequence $(z_t)_{t\geq 0}$ of distinct points of $\mathbb{Z}^2$, with $|z_{t+1} - z_t| \leq D$. A random time is a finitely supported nonnegative integer-valued variable. All later sampling procedures produce such times. The walk itself is never assigned a probability law.

Recall the increment bound

$$K_D = |\{u \in \mathbb{Z}^2 : |u| \leq D\}|.$$

This includes the zero vector; although a self-avoiding walk never uses it as an increment, this harmless upper bound simplifies notation.

Lemma 3.2 (Displacement entropy)**.** *If T_0, T_1 are jointly distributed finite random times and $|T_1 - T_0| \leq n$ almost surely, then*

$$H(z_{T_1} - z_{T_0}) \leq 2\log(n+1) + O_{f,D}(1).$$

The same bound holds conditionally on any event or variable under which the time-difference bound still holds. If F is any finite vector of residue maps, then

$$H(F(z_{T_1})) \geq H(F(z_{T_0})) - H(z_{T_1} - z_{T_0}).$$

Proof. The displacement lies in a disk of radius Dn, which contains at most $C_{f,D}(n+1)^2$ lattice points. Entropy is at most the logarithm of support size. For the last assertion, $F(z_{T_0})$ is determined by $F(z_{T_1})$ and the displacement. The chain rule and subadditivity give the claimed inequality. □

3.4 Continuity and short lists

Lemma 3.3 (Continuity)**.** *If two laws of (X, Y), on the same finite product alphabet, have total variation at most $\epsilon \le 1/2$, and the product alphabet has at most a elements, then*

$$|H(X \mid Y) - H(\widetilde{X} \mid \widetilde{Y})| \le 2h(\epsilon) + 2\epsilon \log a.$$

Proof. A coupling with mismatch probability equal to total variation yields $|H(U) - H(\widetilde{U})| \le h(\epsilon) + \epsilon \log a$ for variables on an alphabet of size at most a. Apply this once to the joint variables and once to their Y-marginals, and subtract. □

The next lemma adapts the error-indicator proof of Fano's inequality to lists.

Lemma 3.4 (A short-list entropy bound)**.** *Let X have at most a values, let Y be finite data, and let $\mathcal{L}(Y)$ be a list of possible values of X. If*

$$\Pr\{X \in \mathcal{L}(Y),\ |\mathcal{L}(Y)| \le b\} = \rho, \qquad 1 \le b \le a,$$

then

$$\log a - H(X \mid Y) \ge \rho \log(a/b) - h(\rho).$$

Proof. Let B indicate the event in the statement. It is determined by (X, Y), so the chain rule gives

$$H(X \mid Y) = H(B \mid Y) + H(X \mid B, Y) \le h(\rho) + \rho \log b + (1 - \rho) \log a.$$

On $B = 1$ the conditional support lies in $\mathcal{L}(Y)$; otherwise the full alphabet bound applies. Rearrangement proves the claim. The event may depend on both X and Y; no independence is used. □

4 Entropy enrichment from a forward segment

Throughout this section, $z_0, z_1, \ldots$ is a fixed self-avoiding walk in $\mathbb{Z}^2$ with $|z_{t+1} - z_t| \le D$, where $D \ge 1$. The walk need not satisfy any congruence restrictions. Our goal is a forward sampling operation that raises the average entropy of residue projections. Two geometric estimates supply the input: a walk segment has many distinct differences, and random root congruences usually separate those differences. We then sample an endpoint of a uniformly chosen distinct difference. Its residue entropy inherits both the entropy already present at the starting point and the entropy created within the segment.

4.1 How geometry produces many differences

Lemma 4.1 (Many differences)**.** *Let E consist of the points of a segment of $n \ge 1$ steps of the walk. Let R be its diameter, choose a line through two points realizing that diameter, and let W be the larger of 1 and the maximum distance of a point of E from that line. Then, for $A = RW$,*

$$cn \le A \le D^2 n^2, \qquad |E - E| \ge c_{f,D} A,$$

where $c > 0$ is absolute and $c_{f,D} > 0$ depends only on D.

Proof. Let p_0, p_1 be the diameter endpoints. Every point of E projects onto the segment p_0p_1: a projection beyond either endpoint would give a distance greater than R to the other endpoint. Thus E lies in a rectangle of length R and width $2W$. Here $R \geq W \geq 1$. Disjoint disks of radius $1/3$ about the distinct lattice points of E lie in a fixed enlargement of this rectangle. Its area is $O(RW)$, proving $n + 1 = O(A)$. The step bound gives $R \leq Dn$ and $W \leq R$, proving the upper bound.

If $W = 1$, subtracting a fixed member of E gives $|E - E| \geq n + 1 \geq A/D$. Suppose instead that $W > 1$, and choose $p_2 \in E$ at distance W from the diameter line. Put

$$a = p_1 - p_0, \qquad b = p_2 - p_0, \qquad \Lambda = \mathbb{Z}a + \mathbb{Z}b.$$

The flat torus $\mathbb{R}^2/\Lambda$ has area $|\det(a, b)| = RW = A$. Follow the interpolated walk, in either time direction as necessary, to obtain paths $\alpha, \gamma : [0, 1] \to \mathbb{R}^2$ from p_0 to p_1, p_2, respectively. The map

$$(s, t) \longmapsto \alpha(s) - \gamma(t) \pmod{\Lambda}$$

descends to a continuous map from the parameter torus $[0, 1]^2/\sim$ to $\mathbb{R}^2/\Lambda$: opposite edges agree because the respective endpoint differences are a and $-b$. Homotoping both paths to their straight segments while keeping their endpoints fixed shows that this map has degree -1, with the target oriented by a, b. In particular it is onto. Indeed, a map omitting a point factors through a punctured torus, whose second homology is zero, and therefore has degree zero.

Every interpolated point is within distance D of a vertex of its walk edge. Consequently the images of $E - E$, with disks of radius $2D$ about them, cover the target torus. Each such projected disk has area at most $4\pi D^2$, regardless of overlap or wrapping around the torus. Hence

$$A \leq 4\pi D^2 |E - E|,$$

as required. □

4.2 Random root kernels avoid thin rectangles

Lemma 4.2 (Rectangle separation)**.** *Let $p_1, \ldots, p_r$ be distinct good primes in $[T, 2T]$, $T \geq 2$, and put $P = \prod_i p_i$. Let Q be an origin-centered rectangle of half-lengths $R_1 \geq W_1 \geq 1$ in any orientation. Suppose*

$$R_1 W_1 \leq P^{1-\eta}, \qquad 0 < \eta < \tfrac{1}{4}.$$

There are positive constants c, C, C_0, depending only on f, such that, if $\eta^3 r \geq C_0$, independent uniform root choices satisfy

$$\Pr\{\exists w \in Q \cap \mathbb{Z}^2 \setminus \{0\} : \ \phi_{p_i,\sigma_i}(w) = 0 \ (1 \leq i \leq r)\} \leq C \exp(-c\eta^3 r).$$

Proof. Call a root-choice vector bad if it admits such a witness. If two choices differ in $h \leq \eta r/4$ coordinates, the product of their common-coordinate primes is at least $P/(2T)^h$. By Lemma 3.1 it divides the determinant of their witnesses. That determinant has absolute value at most $2R_1W_1$, also for a rotated rectangle. The logarithm of the ratio of the divisor to this upper bound is at least

$$\eta r(\log T - \tfrac{1}{4}\log(2T)) - \log 2 > 0$$

when C_0 is large. Thus the witnesses are collinear. This also proves that all witnesses for a single bad choice have one common line.

Fix a line with primitive direction v. Its integer points are exactly av, $a \in \mathbb{Z}$: Bezout's identity applied to the coordinates of v makes the multiplier integral. If a selected coordinate annihilates av but not v, then $p \mid a$. Therefore any nonzero witness has length at least $|v|P/P_\sigma(v)$. Since every point of Q has length at most $\sqrt{2}R_1$ and $R_1 \le P^{1-\eta}$, a witness requires

$$\log P_\sigma(v) \ge \log|v| + \eta \log P - \log\sqrt{2}.$$

The mean on the left is at most $\log|v| + \log C_f/d$. It is a sum of independent variables, each of range at most $\log(2T)$. Hoeffding's bounded-sum inequality consequently gives

$$\Pr\{\text{a witness on this fixed line}\} \le e^{-c_0\eta^2 r}. \tag{4.1}$$

Here C_0 absorbs the fixed coefficient cost, so the required deviation is at least $\eta r \log T/2$; also $\log(2T)/\log T \le 2$. The inequality used is $\Pr\{\sum_i (X_i - \mathbb{E}X_i) \ge t\} \le \exp(-2t^2/\sum_i (b_i - a_i)^2)$ for independent $X_i \in [a_i, b_i]$.

We now combine the line bounds without counting lines. In the Hamming graph $[d]^r$, with an edge for every change of one coordinate, every nonempty E satisfies

$$|\partial_E E| \ge (d-1)|E| \log_d(d^r/|E|). \tag{4.2}$$

To prove this, let X be uniform on E. If the fiber in coordinate i, with the other coordinates fixed, has size k_i, the number of internal neighbors in that coordinate is $k_i - 1$. Concavity of the logarithm gives $k_i - 1 \le (d-1)\log_d k_i$ for $1 \le k_i \le d$. Moreover $\mathbb{E}\log k_i = H(X_i \mid X_{-i})$ and

$$\sum_i H(X_i \mid X_{-i}) \le \sum_i H(X_i \mid X_1, \dots, X_{i-1}) = H(X) = \log|E|.$$

Subtracting the average internal degree from $r(d-1)$ proves (4.2). Each outside vertex has at most $r(d-1)$ incident boundary edges, so the outside vertex boundary has size at least $|E|\log_d(d^r/|E|)/r$.

Partition the bad choices by their witness lines, writing E_ℓ for the nonempty classes. There are finitely many classes because Q contains finitely many lattice points. Distinct classes have distance greater than $\eta r/4$, so their neighborhoods of radius $h = \lfloor \eta r/10 \rfloor$ are disjoint. Each class has density at most $e^{-c_0\eta^2 r}$ by (4.1). Until its growing neighborhood reaches density $e^{-c_0\eta^2 r/2}$, the boundary estimate increases size at each step by a factor at least $1 + c_d\eta^2$, with $c_d > 0$ depending only on d. After h steps the gain is at least $e^{c_d'\eta^3 r}$. If the density threshold was reached earlier, the gain is already at least $e^{c_0\eta^2 r/2}$, which is stronger after reducing c_d'. The disjoint enlarged classes together have at most d^r elements. Their original total density is therefore at most $e^{-c_d'\eta^3 r}$, as required. □

4.3 From differences to endpoint entropy

The two geometric estimates now give a sampling operation that can be iterated. We first specify how entropy is averaged over residue maps; the same convention will be used throughout the sampling schedule.

Fix a batch $\mathcal{P}$ of k distinct good primes in $[T, 2T]$, and write

$$L_* = \frac{1}{k}\sum_{p \in \mathcal{P}} \log p.$$

A root-labelled subset $\mathcal{S}$ consists of a subset of distinct primes from $\mathcal{P}$, with one of its d root coordinates chosen for each. Thus its size counts rational primes, and it never contains two coordinates of the same prime. For a finitely supported point variable $Z \in \mathbb{Z}^2$, define

$$F_{\mathcal{S}}(Z) = (Z \bmod \pi)_{\pi \in \mathcal{S}}, \qquad f_Z(s) = \mathbb{E}_{\mathcal{S}} H(F_{\mathcal{S}}(Z)), \quad 0 \le s \le k,$$

where the s underlying primes are chosen uniformly without replacement and their root labels independently and uniformly. All index choices are independent of the point variables. Entropy is computed for each fixed root-labelled subset before averaging.

We have

$$f_Z(0) = 0, \qquad 0 \le f_Z(s) \le sL_*.$$

The function f_Z is concave on the integer interval. Indeed, choose a uniform random ordering of the batch and independent uniform root labels. The difference $f_Z(s+1) - f_Z(s)$ is the average conditional entropy of the next coordinate given the first s. Exchangeability of the unexposed coordinates and the fact that conditioning decreases entropy show that these differences decrease with s. This is the usual entropy submodularity argument.

Let $Z = z_\tau$, where τ is any finitely supported random time. For parameters $U > 0$ and $0 < g < 10^{-2}$, put $n = \lfloor e^{gU} \rfloor$. For an exact time t, write

$$E_t = \{z_t, z_{t+1}, \dots, z_{t+n}\}.$$

Conditional on $\tau = t$, choose h uniformly from the set $E_t - E_t$, including zero, and then choose one ordered pair $(X, Y) \in E_t^2$ with $X - Y = h$ by a fixed deterministic rule. Thus differences, not ordered pairs, receive equal weight. Finally let V equal X or Y, each with probability $1/2$, using an independent coin. Self-avoidance identifies each endpoint with a unique walk time in $[t, t+n]$, so this defines a forward transition of the time law. The transition advances time by at most n and uses no prime indices.

Lemma 4.3 (Entropy enrichment)**.** *Let $Z = z_\tau$ and V be the starting point and sampled endpoint in the preceding transition for a fixed self-avoiding D-step walk. Set*

$$q = \lfloor U/L_* \rfloor, \qquad b = \lfloor g^2 U/L_* \rfloor.$$

Consider any asymptotic family in which D is fixed, $T \to \infty$, $q \le k$, and

$$\frac{g^C U}{\log T} \longrightarrow \infty \quad \text{for every fixed } C > 0.$$

Uniformly over the walks, batches, and starting-time laws,

$$\frac{f_V(b)}{bL_*} \ge \frac{1}{2}\left(1 + \frac{f_Z(q)}{qL_*}\right) - O_{f,D}(g). \tag{4.3}$$

The right side retains one half of the starting entropy fraction and adds one half of the maximal fraction. When successive transitions match the next input coordinate count to the preceding output count, repetition therefore drives the entropy deficit downward. Section 6 will arrange this matching, starting even from a deterministic time. The error $O_{f,D}(g)$ is uniform; choosing the scale large enough permits g to decrease as well.

Proof. The regime ensures $1 \le b < q$ for sufficiently large parameters. We first show that a fresh difference coordinate has nearly maximal entropy even after the endpoint prefixes are known. We then use the entropy chain rule to retain the contribution from Z.

Entropy of one further difference coordinate. Choose a uniform root-labelled set $\mathcal{Q}$ of size q, and within it a uniform root-labelled prefix $\mathcal{S}$ of size b. Write

$$B_x = F_{\mathcal{S}}(X), \qquad B_y = F_{\mathcal{S}}(Y), \qquad h = X - Y.$$

If j is a uniform remaining root coordinate and $h_j = F_j(h)$, we first prove

$$\mathbb{E}_{\mathcal{Q},\mathcal{S},j} H(h_j \mid B_x, B_y) \ge (1 - O_{f,D}(g))L_*. \tag{4.4}$$

Fix an exact base time $\tau = t$ with positive probability, and use A from Lemma 4.1 for its segment. That lemma gives

$$gU - O_{f,D}(1) \le \log A \le 2gU + O_{f,D}(1).$$

Put

$$r = \left\lceil (1 + 10g)\frac{\log A}{L_*} \right\rceil.$$

For sufficiently large parameters, $gq/2 \le r \le 3gq \le q - b$. Choose a uniform size-$r$ subset $\mathcal{J}$ of the remaining coordinates. Averaging over $\mathcal{Q}, \mathcal{S}$, its underlying primes form a uniform sample of size r from the batch. Their log-product has mean rL_* and variance $O(r)$, since the values $\log p$ lie in an interval of length $\log 2$ and sampling is without replacement. Hence

$$\Pr\left\{ \sum_{p \in \mathcal{J}} \log p < (1 + 5g) \log A \right\} = O\left(\frac{1}{g^2 r L_*^2} \right). \tag{4.5}$$

In coordinates along the diameter rectangle, differences of differences from the segment lie in an origin-centered rectangle of half-lengths $4R, 4W$, whose product is $16A$. On the complementary event in (4.5), writing $P = \prod_{p \in \mathcal{J}} p$, we have

$$P^{1-g} \ge A^{(1+5g)(1-g)} \ge A^{1+3g} > 16A$$

for sufficiently large parameters. Lemma 4.2, with $\eta = g$, shows that reduction in the coordinates $\mathcal{J}$ is injective on the difference set, except with additional probability $O(e^{-cg^3 r})$. Let

$$\epsilon = O\left(\frac{1}{g^2 r L_*^2} + e^{-cg^3 r} \right)$$

be the total exceptional probability.

Here all errors are uniform in the exact base time. Indeed r is bounded above and below by constant multiples of gU/L_*, while L_* is comparable to $\log T$. The assumed regime implies

$$\frac{1}{gq} = o(g), \qquad \frac{1}{g^2 r L_*^2} = o(g), \qquad g^3 r / \log(1/g) \longrightarrow \infty$$

when $g \to 0$; for g bounded away from zero the last assertion is replaced simply by $g^3 r \to \infty$. For example, the condition with $C = 5$ makes $g^3 r$ larger than an unbounded multiple of $1/g$. Thus $\epsilon = o(g)$, ceiling errors are $o(g)$ relatively, and $g \log A \to \infty$, as used above.

On a good choice of $\mathcal{J}$, uniformity on the distinct differences and injectivity give

$$H(F_{\mathcal{J}}(h) \mid \tau = t) \geq \log A - O_{f,D}(1).$$

Conditioning additionally on B_x, B_y costs at most $2b\log(2T)$. Averaging the fixed-index entropies therefore yields

$$\mathbb{E}_{\mathcal{Q},\mathcal{S},\mathcal{J}} H(F_{\mathcal{J}}(h) \mid B_x, B_y, \tau = t) \geq (1-\epsilon)(\log A - O_{f,D}(1)) - 2b\log(2T).$$

For every fixed index choice, conditional subadditivity gives

$$H(F_{\mathcal{J}}(h) \mid B_x, B_y, \tau = t) \leq \sum_{j \in \mathcal{J}} H(h_j \mid B_x, B_y, \tau = t).$$

Consequently,

$$\frac{1}{r}\mathbb{E}_{\mathcal{Q},\mathcal{S},\mathcal{J}} H(F_{\mathcal{J}}(h) \mid B_x, B_y, \tau = t) \leq \mathbb{E}_{\mathcal{Q},\mathcal{S},j} H(h_j \mid B_x, B_y, \tau = t),$$

where j is uniform among the remaining coordinates. This remains true although r depends on t: at this stage t is fixed, and every member of the uniform $\mathcal{J}$ has the same uniform remaining-coordinate marginal. Since

$$\frac{\log A}{rL_*} \geq 1 - O(g), \qquad \frac{b}{r} = O(g),$$

we obtain the right side of (4.4), still with the extra conditioning $\tau = t$. Averaging over t, and then removing the conditioning on τ, proves (4.4).

Retaining the starting entropy. We have established near-maximal entropy for a further coordinate of $h = X - Y$, after revealing the endpoint prefixes. It remains to turn that information into endpoint entropy while keeping the contribution from the starting law. For fixed index choices, the pair $(F_j(X), F_j(Y))$ is equivalent to $(h_j, F_j(X))$. The entropy chain rule and conditioning inequalities give

$$\begin{aligned} H(h_j \mid B_x, B_y) &= H(F_j(X), F_j(Y) \mid B_x, B_y) - H(F_j(X) \mid B_x, B_y, h_j) \\ &\leq H(F_j(X) \mid B_x) + H(F_j(Y) \mid B_y) - H(F_j(X) \mid B_x, B_y, h). \end{aligned} \tag{4.6}$$

In particular the last conditioning is on the entire difference h, not just on h_j; using it decreases the subtracted entropy and preserves the displayed inequality. Summing these subtracted entropies over the $q - b$ remaining coordinates gives at least

$$H(F_{\mathcal{Q}}(X) \mid B_x, B_y, h),$$

because the prefix coordinates are already known from B_x. Its average is at least

$$f_X(q) - 2b\log(2T) - H(h).$$

Also $F_{\mathcal{Q}}(Z)$ is determined by $F_{\mathcal{Q}}(X)$ and $X - Z$, so

$$f_X(q) \geq f_Z(q) - H(X - Z).$$

Both h and $X - Z$ are lattice displacements of length at most Dn. The counting bound of Lemma 3.2 gives

$$H(h) + H(X - Z) \leq 4\log(n+1) + O_{f,D}(1) = O_{f,D}(gU).$$

Consequently the average contribution of the subtracted term in (4.6) is at least

$$\frac{f_Z(q) - O_{f,D}(gU)}{q-b} \geq \frac{f_Z(q)}{q} - O_{f,D}(g)L_*.$$

The averages of the first two terms on the right side of (4.6) are, respectively, $f_X(b+1)-f_X(b)$ and $f_Y(b+1)-f_Y(b)$. Combining with (4.4) proves

$$[f_X(b+1)-f_X(b)] + [f_Y(b+1)-f_Y(b)] \geq L_* + \frac{f_Z(q)}{q} - O_{f,D}(g)L_*.$$

Concavity and $f_X(0) = f_Y(0) = 0$ bound these forward differences above by $f_X(b)/b$ and $f_Y(b)/b$. Finally, concavity of entropy under a mixture gives $f_V(b) \geq \frac{1}{2}(f_X(b) + f_Y(b))$. Dividing by bL_* proves (4.3). □

5 Coverage from shared continuation data

The geometric argument provides joint residue entropy. Before choosing a schedule of geometric operations, we prove the coverage principle that will determine what the schedule must supply. The principle uses only finite abelian residue groups and applies to the root coordinates just defined.

5.1 One shared conditioning cost

We first isolate why a common data vector is useful. If one separately conditioned on data chosen for each coordinate, its entropy cost would be paid once per coordinate. A single vector can instead be paid for once in the joint entropy, then distributed over the coordinates.

Let Λ be a lattice, that is, a finitely generated free abelian group, and let $k \geq 1$ be an integer. For $1 \leq j \leq k$ and $\sigma \in \{1, \dots, d\}$ let

$$\phi_{j,\sigma} : \Lambda \longrightarrow G_{j,\sigma}$$

be homomorphisms to finite abelian groups of order p_j. Assume $T \leq p_j \leq 2T$ and $T \geq 2$, and put

$$\overline{L} = \frac{1}{k}\sum_{j=1}^{k} \log p_j, \qquad \operatorname*{av}_c u_c = \frac{1}{dk}\sum_{j=1}^{k}\sum_{\sigma\in\{1,\dots,d\}} u_{j,\sigma}.$$

Here $c = (j,\sigma)$ denotes one root coordinate; write $p_c = p_j$, $G_c = G_{j,\sigma}$, and $\phi_c = \phi_{j,\sigma}$. For an integer $1 \leq s \leq k$, choose s distinct indices uniformly and one independent uniform root label at each. Denote this root-labelled subset by S, and write $F_S(Y) = (\phi_c(Y))_{c\in S}$ for a finite Λ-valued variable Y. The choice of S is auxiliary and independent of every variable whose entropy is measured. In particular, $\mathbb{E}_S H(F_S(Y))$ means the average of the entropies for fixed S.

Lemma 5.1 (One shared conditioning cost)**.** *Let $(Y, \mathcal{H})$ be any pair of finite variables, with Y taking values in Λ, and let S be selected independently as above. If*

$$\mathbb{E}_S H(F_S(Y)) \geq (1-\epsilon)s\overline{L}, \qquad H(\mathcal{H}) \leq \kappa s\overline{L},$$

then

$$\operatorname*{av}_c [\log p_c - H(\phi_c(Y) \mid \mathcal{H})] \leq (\epsilon + \kappa)\overline{L}. \tag{5.1}$$

No independence between Y and $\mathcal{H}$ is required.

Proof. For each fixed S, the chain rule and subadditivity give

$$H(F_S(Y)) - H(\mathcal{H}) \leq H(F_S(Y) \mid \mathcal{H}) \leq \sum_{c \in S} H(\phi_c(Y) \mid \mathcal{H}).$$

Each root coordinate belongs to S with probability $s/(dk)$. Averaging and dividing by s shows that the mean conditional entropy is at least $(1-\epsilon-\kappa)\overline{L}$. The mean alphabet entropy is $\overline{L}$, which proves the assertion. □

We will contradict (5.1) with the short-list bound of Lemma 3.4. Its success event may depend on both the residue and the data; independence of these variables is unnecessary.

5.2 A transfer from later coverage

We now mix a family of displacement laws. Each conditional law will have a probability floor outside a small exceptional set. High joint entropy of the starting position, together with a small cost for shared data, will strengthen the exceptional-set bound for the mixture.

Let A be a finite variable, let y map its alphabet to Λ, and put $Y = y(A)$. For every value a of A, specify a finitely supported displacement law ν_a on Λ. Conditional on A, sample $h_1, \ldots, h_n$ independently with law ν_A, where $n \geq 1$ is an integer, and define

$$\mathcal{H} = (h_1, \ldots, h_n), \qquad Z = Y + h_1.$$

In the walk application, A will record an exact intermediate time, Y its position, and ν_a the displacement law from that time to the terminal position. Here A is simply the finite mixing variable; its influence on both Y and $\mathcal{H}$ is retained. The coordinate subset S used to measure entropy is independent of this whole experiment.

For a variable X in an alphabet of size p, say explicitly that it has (τ, β) coverage when

$$\#\{u : \mathbb{P}(X = u) < \tau/p\} < p^{1-\beta}. \tag{5.2}$$

The strict inequality concerns the number of exceptional residues; no claim is made that this set is empty.

The transfer below lowers the probability floor from τ' to τ, but improves the exceptional-set bound from $p^{1-\beta'}$ to $p^{1-\beta}$, with $\beta \geq \beta'$. The exceptional sets of the conditional laws may vary with a, so mixing them does not directly give this improved bound. The theorem asks for three inputs: coverage for each conditional law, a joint entropy budget that can absorb the common data, and enough independent trials to make the candidate lists small.

Theorem 5.2 (One-step coverage transfer)**.** *Use the root coordinates and the experiment just defined, with integers $1 \le s \le k$ and $n \ge 1$. Let*

$$0 < \tau \le 3/8, \quad 0 < \delta \le 1/64, \quad \tau' = \tau + 4\delta, \quad 0 < \beta' \le \beta < 1, \quad 0 < \eta \le 1, \quad 0 \le \eta' \le 1.$$

Suppose that at every a in the support of A, all but a fraction at most η' of the root coordinates have (τ', β') coverage for $\phi_c(y(a) + h_1)$ under the law conditional on $A = a$. Assume also

$$\mathbb{E}_S H(F_S(Y)) \ge (1-\epsilon)s\overline{L}, \qquad H(\mathcal{H}) \le \kappa s\overline{L}, \tag{5.3}$$

$$\eta' \le \eta\delta/100, \qquad \epsilon + \kappa \le \eta\delta\beta'/16, \tag{5.4}$$

where $\epsilon, \kappa \ge 0$. Finally, for every coordinate size $p = p_j$, assume

$$p\exp(-\delta^2 np^{-\beta}/2) \le \delta/100, \qquad \delta^{-1} < p^{\beta'/2}, \tag{5.5}$$

$$\log(2e/\delta) \le (\beta'/4)\log T. \tag{5.6}$$

Then all but a fraction at most η of the coordinates have (τ, β) coverage for $\phi_c(Z)$.

Proof. Call a coordinate deficient if its terminal law $\phi_c(Z)$ fails (5.2). Suppose their fraction is $\theta > \eta$. For each deficient coordinate of size p choose

$$K_c \subseteq \{u : \mathbb{P}(\phi_c(Z) = u) < \tau/p\}, \qquad |K_c| = u_p := \lceil p^{1-\beta} \rceil.$$

Such a set exists, and $1 \le u_p \le p$. Using the same data $\mathcal{H}$ for all coordinates, define

$$\mathcal{L}_c(\mathcal{H}) = \left\{ x \in G_c : \sum_{\ell=1}^{n} \mathbf{1}_{\{x+\phi_c(h_\ell) \in K_c\}} \le n(\tau+\delta)u_p/p \right\}. \tag{5.7}$$

We will show that the true residue belongs to this list often, but that the list is small on average over deficient coordinates.

The true residue is often on the list. Every $Y + h_\ell$ has the same law as Z. The expectation of the count in (5.7) at $x = \phi_c(Y)$ is therefore at most $n\tau u_p/p$. Markov's inequality gives

$$\mathbb{P}\{\phi_c(Y) \in \mathcal{L}_c(\mathcal{H})\} \ge 1 - \frac{\tau}{\tau+\delta} = \frac{\delta}{\tau+\delta} \ge \delta. \tag{5.8}$$

This step uses only the common one-trial marginal, not independence of the trials before conditioning on A.

Coverage at the next checkpoint makes the list small. Fix $A = a$ and a coordinate whose conditional terminal law has (τ', β') coverage. Translation by $\phi_c(y(a))$ shows that the conditional law of $\phi_c(h_\ell)$ gives probability at least τ'/p outside a set $E_{a,c}$ of size less than $p^{1-\beta'}$. In any finite abelian group,

$$\sum_{x \in G_c} |(K_c - x) \cap E_{a,c}| = u_p |E_{a,c}|.$$

Hence, except for at most $|E_{a,c}|/\delta$ candidates x,

$$|(K_c - x) \cap E_{a,c}| \le \delta u_p.$$

For every other x, each conditional trial has hit probability at least

$$\tau'(1-\delta)u_p/p \geq (\tau+2\delta)u_p/p =: \lambda.$$

Indeed $(\tau+4\delta)(1-\delta)-(\tau+2\delta) = \delta(2-\tau-4\delta) > 0$.

The count for such an x stochastically dominates $\mathrm{Bin}(n,\lambda)$, because its trials are independent conditional on this exact a. The multiplicative lower-tail estimate gives

$$\mathbb{P}\{\mathrm{Bin}(n,\lambda) \leq n(\lambda-\delta u_p/p)\} \leq \exp\left(-\frac{n(\delta u_p/p)^2}{2\lambda}\right) \leq \exp(-\delta^2 np^{-\beta}/2).$$

For completeness, if $B \sim \mathrm{Bin}(n,\lambda)$ and $0 < v < 1$, then $\mathbb{E}e^{-tB} \leq \exp(n\lambda(e^{-t}-1))$. Markov's inequality with $t = -\log(1-v)$ bounds $\mathbb{P}\{B \leq (1-v)n\lambda\}$ by $\exp[-n\lambda(v+(1-v)\log(1-v))] \leq \exp(-n\lambda v^2/2)$. Apply this with $v = \delta/(\tau+2\delta)$; the last inequality in the displayed bound uses $u_p/p \geq p^{-\beta}$ and $\tau+2\delta < 1$.

A union bound over the at most p candidates now shows that, with conditional probability at least $1-\delta/100$, the list contains only the exceptional candidates. By (5.5), their number is less than

$$\delta^{-1}p^{1-\beta'} < p^{1-\beta'/2}.$$

In particular,

$$\mathbb{P}\{|\mathcal{L}_c(\mathcal{H})| > p^{1-\beta'/2} \mid A = a\} \leq \delta/100 \tag{5.9}$$

whenever this coordinate has the required coverage at a.

We have used next-checkpoint coverage only after fixing its exact value. It remains to average over that value without assuming that a coordinate deficient now is necessarily covered at the next checkpoint.

Averaging yields too much conditional information. At each a, the fraction of coordinates without the required coverage is at most η'. Averaging first over A and then over the θ fraction of deficient coordinates, the mean probability of such a failure is at most

$$\eta'/\theta < \delta/100.$$

Let

$$a_c = \mathbb{P}\{\phi_c(Y) \in \mathcal{L}_c(\mathcal{H}), \quad |\mathcal{L}_c(\mathcal{H})| \leq p_c^{1-\beta'/2}\}.$$

Equations (5.8) and (5.9) imply that its mean $\overline{a}$ over the deficient coordinates satisfies

$$\overline{a} \geq \delta - \delta/100 - \eta'/\theta \geq \delta/2.$$

Lemma 3.4 gives, for each of these coordinates,

$$\log p_c - H(\phi_c(Y) \mid \mathcal{H}) \geq a_c(\beta'/2)\log p_c - h(a_c).$$

Concavity of h and $h(u) \leq u\log(e/u)$ give

$$\operatorname*{av}_{c \text{ deficient}} h(a_c) \leq h(\overline{a}) \leq \overline{a}\log(2e/\delta).$$

After averaging and applying (5.6), we obtain

$$\operatorname*{av}_{c \text{ deficient}} [\log p_c - H(\phi_c(Y) \mid \mathcal{H})] \geq \frac{\overline{a}\beta'}{4}\log T \geq \frac{\delta\beta'}{8}\log T.$$

Every other conditional entropy deficit is nonnegative. Since $\overline{L} \le \log(2T) \le 2\log T$ and $\theta > \eta$, the mean over all coordinates is strictly greater than

$$\frac{\eta\delta\beta'}{8}\log T \ge \frac{\eta\delta\beta'}{16}\overline{L}.$$

This contradicts Lemma 5.1 and (5.4). Thus $\theta \le \eta$. □

The theorem has charged the repeated displacements once, through $H(\mathcal{H})$. Its independence assumption is only conditional on the exact checkpoint A. In the next section the schedule will make the joint entropy error and this single data cost small enough to transfer coverage at every step of a backward induction.

6 A common sampling schedule

The transfer theorem explains what a useful schedule must achieve. At each checkpoint, the joint residue entropy must be almost maximal, while the entropy of repeated continuations must be small compared with that joint entropy. A much more accurate marginal estimate is needed only at the last checkpoint, where it starts the backward coverage induction. We now construct one schedule that supplies these estimates for every prime batch.

The same construction also supplies the inputs needed after coverage has been established: a persistent joint entropy of order T for each batch near T, a smaller total alphabet budget for all preceding batches, and one common terminal-time law whose small translates are close in total variation. These requirements have different roles. The displacement bound pays for information revealed in the coverage argument; the final translation bound will permit the information costs of different batches to be added.

Throughout the section, the walk is an arbitrary fixed infinite self-avoiding D-step walk. No zero-avoidance assumption is used. We first give the actual schedule and prove its entropy and displacement estimates, then apply Theorem 5.2 to obtain coverage.

6.1 Choice of constants and scales

Fix $D \ge 1$, and set

$$\beta_0 = \frac{1}{200d}, \qquad e_0 = \frac{\beta_0}{50}.$$

We will first choose a fixed iteration count l_0 and a sufficiently small constant $0 < g_0 < 10^{-2}$, depending only on f, D, e_0. They determine a constant $c_0 > 0$; only then do we choose a sufficiently large fixed $B > 2$. The precise requirements are specified below. All these constants are fixed before the asymptotic parameter M, and are independent of the walk and starting time.

Let M be a positive integer tending to infinity, and put

$$X_m = (100d)^m \quad (\lceil M/2 \rceil \le m \le M).$$

Use every batch $\mathcal{B}_T$ with

$$T = B^j, \quad j \in \mathbb{Z}, \qquad X_m \le \log T \le 1.05X_m \quad \text{for some } \lceil M/2 \rceil \le m \le M. \tag{6.1}$$

The batches are disjoint because $B > 2$. Write $T_+ = \exp(1.05X_M)$. For this value of M, the union of all these batches is the finite prime selection that will be used in Theorem 1.2.

Starting at time 0 on a fixed infinite self-avoiding D-step walk, apply the transition of Lemma 4.3 with scales U in decreasing order. For each m, use the three bands in Table 1. In a band with parameter g, start at its upper endpoint in $\log U$, and successively subtract $2\log(1/g)$ as long as the resulting scale stays in that band. Equivalently, successive U's have ratio g^2. All randomizations are fresh conditional on the current exact walk time. No prime subset or root choice enters these transitions.

Band	Interval for $\log U$	Transition parameter
Top	$[0.8X_m, 1.3X_m]$	g_0
Middle	$[0.20X_m/d, 0.30X_m/d]$	$g_1 = e^{-M^3}$
Bottom	$[0.05X_m/d, 0.10X_m/d]$	$g_2 = e^{-100M^{20}}$

Table 1: The three bands in each scale window. Windows are processed in decreasing m, and scales decrease within each band.

After all transitions in all windows have been performed, add one independent uniform offset in $\{0, \dots, N-1\}$, where $N = \lceil T_+^{10} \rceil$. Denote the resulting random time by t_*. Thus all batches will be measured at the same time law.

A *checkpoint* is a stage just before a specified transition, or the terminal stage. It also denotes the random walk time at that stage. Conditioning on an exact start at a checkpoint means fixing that walk time and then running the remaining schedule with fresh randomness. Such conditional experiments are defined at every nonnegative integer start, whether or not it has positive probability in the experiment starting from time 0. The term "exact start" specifies a new continuation experiment, not conditioning on an event that may have probability zero.

Lemma 6.1 (Future shifts and smoothing)**.** *Let V be a scheduled scale in a band with parameter g. For all sufficiently large M, the maximum total forward time shift from the checkpoint before V to any later checkpoint, including t_*, is bounded by a deterministic number S_V satisfying*

$$\log(1 + S_V) \le 2gV.$$

Consequently the entropy of its physical displacement is $O_{f,D}(gV)$, for every starting law. Furthermore, conditional on any earlier exact checkpoint start,

$$\mathrm{TV}\big(\mathcal{L}(t_*), \mathcal{L}(t_* + a)\big) \le T_+^{-9} \qquad (0 \le a \le T_+,\ a \in \mathbb{Z}). \tag{6.2}$$

Proof. One may take the deterministic bound

$$S_V = N - 1 + \sum_{U \text{ remaining}} \lfloor \exp(g_U U) \rfloor,$$

where the sum includes the transition at V and g_U is the parameter of the band containing U. Every summand and the offset length N are fixed by the schedule, before any exact start is chosen. This bounds the total advance from every such start. The number of operations is $O_{g_0}(X_M) = \exp(O_{g_0}(M))$. A transition at scale U shifts time by at most $\exp(gU)$. Within the same band all future log-lengths are at most gV. Between successive bands the decrease in U is at least $\exp(cX_m)$ for a constant $c = c(d) > 0$, including the passage from the bottom band for m to the top band for $m-1$: $0.05X_m/d - 1.3X_{m-1} = 0.037X_m/d$. This decrease dominates every

ratio of transition parameters, because $\log(1/g_2) = 100M^{20} = o(X_{\lceil M/2 \rceil})$. Thus later bands also have log-length at most gV for large M.

The logarithm of the final offset length is $O(X_M)$. On the other hand, for the least possible scale and parameter,

$$\log(gV) \geq 0.05 X_{\lceil M/2 \rceil}/d - 100M^{20},$$

so $X_M = o(gV)$, uniformly. Taking the logarithm of the sum of all remaining lengths adds only $O_{g_0}(M)$. The bound $2gV$ follows, and Lemma 3.2 gives the entropy assertion.

Write $t_* = \widetilde{t} + U_N$, where U_N is the final uniform offset, independent of all earlier choices. A translate by $a \leq T_+$ changes its law by at most a/N. Mixtures and deterministic translations do not increase this bound, including after an earlier exact start is fixed. This proves (6.2). □

6.2 Iterating and preserving joint entropy

For a fixed batch write $f_Z(s)$ as in Section 4. Every use of Lemma 4.3 below has $q \leq k_T$. For each entropy estimate we begin at a scale where $q \leq k_T$; all subsequent scales in that block are smaller. The other limiting conditions hold uniformly:

$$\log U \geq c X_{\lceil M/2 \rceil}, \qquad \log(1/g) = O(M^{20}), \qquad \log\log T = O(M).$$

In particular $g^C U/\log T \to \infty$ for every fixed $C > 0$.

Suppose l successive transitions in one band end just before its next scale V, and that the initial subset size is at most k_T. The output size of one transition and the input size of the next are both exactly $\lfloor g^2 U/L_* \rfloor$. Writing the normalized entropy deficit as d_j, Equation (4.3) gives $d_{j+1} \leq d_j/2 + C_{f,D} g$. Since $d_0 \leq 1$, iteration gives $d_l \leq 2^{-l} + 2C_{f,D} g$, and hence

$$\frac{f_Z(\lfloor V/L_* \rfloor)}{\lfloor V/L_* \rfloor L_*} \geq 1 - 2^{-l} - O_{f,D}(g) \tag{6.3}$$

at that checkpoint. The same bound, with another $O_{f,D}(g)$ error, holds at every later checkpoint. Indeed Lemmas 6.1 and 3.2 show that the entropy loss at this fixed subset size is $O_{f,D}(gV)$, whereas $\lfloor V/L_* \rfloor L_* \geq V/2$ for large M. The argument is uniform for every initial law, including an exact start fixed before these l transitions. The later entropies here are marginal entropies in that experiment; we are not conditioning on the exact later time, which would determine its walk position.

Choose a small constant $a_0 = a_0(f) > 0$ such that $a_0 T/L_* \leq k_T$ for large T, using (3.2). For a batch T in window m, begin an entropy estimate at the first top-band scale $U \leq a_0 T$. Its size lies between $g_0^2 a_0 T$ and $a_0 T$. Choose a fixed integer l_0 large enough that $2^{-l_0} < e_0/3$, and take g_0 small enough to make the total error in (6.3) and its preservation less than $2e_0/3$. These choices can be made before B and M. Run l_0 transitions and let $q_T = \lfloor V/L_* \rfloor$ be the recorded size. All these scales lie in the top band for large M. At every later checkpoint, for its point variable Z,

$$f_Z(q_T) \geq (1 - e_0) q_T L_*, \qquad q_T L_* \geq c_0 T \tag{6.4}$$

with $c_0 > 0$ depending only on the fixed choices. For example any sufficiently small constant below $a_0 g_0^{2(l_0+1)}/2$ works in the second inequality.

By (3.2), $k_{T'} \log(2T') \leq CT'$. Since the batches are a subsequence of the geometric progression B^j, we can now choose $B > 2$ so large that for every used batch

$$\sum_{T' < T} k_{T'} \log(2T') \leq e_0 c_0 T. \tag{6.5}$$

In fact the left side is at most $CT/(B-1)$. This completes the fixed-constant choices in the order f, D, e_0, then l_0, g_0, c_0, then B, and finally M. The top band has now supplied a batch entropy of order T that survives the remaining schedule, and the geometric spacing makes all smaller residue alphabets inexpensive compared with it. We next build the more accurate conditional estimates needed for coverage.

6.3 Intermediate checkpoints and almost uniform marginals

These checkpoints will transmit coverage backward from an accurate one-coordinate base case. Their spacing must allow joint entropy to grow while keeping subsequent displacement data inexpensive. For each batch T, put

$$\beta_i = 2^{-i}\beta_0, \qquad r_0 = \tfrac{1}{4d}, \qquad r_{i+1} = r_i - 2\beta_i. \tag{6.6}$$

Stop at the least $J = J(T)$ with $\beta_J \log T \le M^{20}$. Since $\beta_0 \log T > M^{20}$ for sufficiently large M, we have $J \ge 1$; minimality then gives

$$J = O(M), \qquad M^{20}/2 < \beta_J \log T \le M^{20}, \qquad 0.23/d < r_i \le 0.25/d. \tag{6.7}$$

The last assertion follows from $r_i = 1/(4d) - 4\beta_0(1 - 2^{-i})$. Let A_i be the checkpoint just before the first scheduled operation with $U \le T^{r_i}$. All these checkpoints lie in the middle band for this m, since

$$0.23X_m/d < r_i \log T \le 0.2625X_m/d.$$

Its scale is in $(g_1^2 T^{r_i}, T^{r_i}]$. Let S_i be a deterministic upper bound for the remaining time shift from A_i supplied by Lemma 6.1, and define

$$\mathcal{D}_i = \{v \in \mathbb{Z}^2 : |v| \le DS_i\}.$$

This set does not depend on the exact start. From every exact start $A_i = a$, every possible displacement to any later checkpoint, including $z_{t_*} - z_a$, belongs to $\mathcal{D}_i$. Since the scale at A_i is at most T^{r_i}, lattice-point counting gives

$$\log |\mathcal{D}_i| \le C_{f,D} T^{r_i}. \tag{6.8}$$

Consequently, uniformly in the starting law, for any later checkpoint A we have

$$H(z_A - z_{A_i}) \le C_{f,D} T^{r_i}. \tag{6.9}$$

The common support bound is stronger than a separate conditional entropy bound at each exact start. It will control the unconditional entropy of displacement data when the next checkpoint is random.

Fix an exact start at A_i, where $i < J$, and let $Y = z_{A_{i+1}}$. There is a deterministic subset size s_i, depending on the batch and the schedule but not on that start, such that

$$f_Y(s_i) \ge (1 - e^{-cM^3}) s_i L_*, \qquad s_i L_* \ge \exp(r_i \log T - O(M^6)) \tag{6.10}$$

for a fixed $c > 0$. To see this, take $\lceil M^3 \rceil$ iterations starting at A_i. The first log-scale differs from $r_i \log T$ by less than $2M^3$, and these iterations consume $O(M^6)$ log-scale. The starting subset size is at most k_T, since $U \le T^{r_i} \le T^{1/(4d)} = o(T)$ and $k_T L_* \asymp T$. They fit before A_{i+1}, because

$$(r_i - r_{i+1}) \log T = 2\beta_i \log T > 2M^{20}.$$

Equation (6.3), preservation, and $g_1 = e^{-M^3}$ prove the first bound, and the recorded scale proves the second.

Finally, conditional on any exact start at A_J, proceed to the bottom band for this m. Take $\lceil 200M^{20} \rceil$ iterations, recording a subset size s. Begin at its first scale, which is at most $\exp(0.10X_m/d) = o(T)$, so again the initial subset size is at most k_T. The required log-scale span is $O(M^{40}) = o(X_m)$, so the iterations and their recording scale fit in that band. Their deficit, including preservation to t_*, is at most

$$2^{-\lceil 200M^{20} \rceil} + O_{f,D}(e^{-100M^{20}}) \le e^{-90M^{20}}.$$

Concavity of f gives $f(1) \ge f(s)/s$. Thus in every such exact-start experiment,

$$f_{z_{t_*}}(1) \ge (1 - e^{-90M^{20}})L_*. \tag{6.11}$$

Equations (6.10) and (6.11) are the two inputs to the coverage induction: high joint entropy between checkpoints and an extremely accurate one-coordinate base case. Sampling proceeds forward through all transitions; the proof of coverage will proceed backward from A_J to A_0. Every continuation itself still moves forward along the walk.

6.4 Coverage from every exact checkpoint

We now apply the earlier transfer theorem to the estimates just proved. The coordinate groups are $\mathbb{F}_p$, their maps are reduction in the d root directions over each prime of $\mathcal{B}_T$, and the abstract average $\overline{L}$ is exactly L_*. Averages over root coordinates assign equal weight to all dk_T prime-root pairs.

Recall that (τ, β) coverage means that a law on an alphabet of size p assigns mass at least τ/p outside fewer than $p^{1-\beta}$ residues. We use the following floors and exceptional fractions at the checkpoints already constructed:

$$\tau_0 = \tfrac{1}{4}, \qquad \delta_i = 2^{-i-6}, \qquad \tau_{i+1} = \tau_i + 4\delta_i, \tag{6.12}$$

$$\eta_0 = \tfrac{1}{100}, \qquad \eta_{i+1} = \eta_i \delta_i / 100. \tag{6.13}$$

Here β_i, r_i, and J are those of (6.6)–(6.7). In particular,

$$\tau_i = \frac{3}{8} - 2^{-i-3} < \frac{3}{8}, \qquad \eta_i = 100^{-(i+1)} 2^{-i(i-1)/2 - 6i}.$$

Since $J = O(M)$, these formulas imply, uniformly over $0 \le i \le J$,

$$\log(1/\delta_i) = O(M), \qquad \eta_i \ge \exp(-O(M^2)), \qquad \eta_i \delta_i \beta_{i+1} \ge \exp(-O(M^2)). \tag{6.14}$$

The last estimate is needed only for $i < J$.

For a specified exact start a at checkpoint A_i, denote the terminal residue law of its continuation experiment by

$$b_{i,a,\pi}(u) = \Pr\{z_{t_*} \bmod \pi = u : \text{the remaining schedule is started at } A_i = a\}.$$

The colon specifies the experiment, including at starts that have probability zero under the schedule begun at time 0.

Proposition 6.2 (Terminal residue coverage)**.** *For all sufficiently large M, uniformly over the batches, $0 \le i \le J$, and all exact start times $a \ge 0$ at A_i, all but a fraction at most η_i of the root coordinates satisfy*

$$\#\{u \in \mathbb{F}_p : b_{i,a,\pi}(u) < \tau_i/p\} < p^{1-\beta_i}. \tag{6.15}$$

Proof. We begin at A_J using the particularly accurate marginal estimate, then verify every numerical hypothesis of Theorem 5.2 for the passage from A_{i+1} to A_i.

The terminal base. Fix any exact start at A_J. By (6.11), the average relative entropy of the terminal residue laws from their respective uniform laws is at most $e^{-90M^{20}}L_*$. If a root coordinate fails (6.15), its low-probability set has at least $p^{1-\beta_J}$ elements. The uniform law assigns that set at least $(1-\tau_J)p^{-\beta_J}$ more mass than its terminal law. Thus the total variation distance is at least

$$(1-\tau_J)p^{-\beta_J} \ge ce^{-M^{20}},$$

using $p \le 2T$, $\tau_J < 3/8$, and $\beta_J \log T \le M^{20}$. Pinsker's inequality in natural logarithms makes each such entropy deficit at least $c'e^{-2M^{20}}$. Averaging the lower bound over the failing root coordinates bounds their fraction by

$$CL_*e^{-88M^{20}} = \exp(-88M^{20} + O(M)) < \eta_J$$

for all sufficiently large M. This proves the base at every exact start. The terminal marginal deficit has the accuracy needed to absorb the power-saving exceptional-set threshold.

The shared data and its cost. Suppose coverage has been proved at $i+1$, and fix an arbitrary exact start $A_i = a$. Run the schedule to the next checkpoint and put $A = A_{i+1}$ and $Y = z_A$. Conditional on this exact time A, repeat the remaining continuation independently

$$n_i = \left\lceil T^{6\beta_i/5} \right\rceil$$

times. Write $h_1, \dots, h_{n_i}$ for their physical displacements from Y, and $\mathcal{H} = (h_1, \dots, h_{n_i})$. Every $Y + h_\ell$ has the terminal law from the fixed start $A_i = a$. This one data vector is constructed before any auxiliary root coordinate subset is selected.

By (6.8), every h_ℓ belongs to the same finite set $\mathcal{D}_{i+1}$, even as A varies. Hence its *unconditional* entropy in this fixed-start experiment satisfies

$$H(\mathcal{H}) \le n_i \log |\mathcal{D}_{i+1}| \le C_{f,D} n_i T^{r_{i+1}}.$$

This is not an assertion of independence before conditioning on A. Combining the support bound with (6.10) yields

$$\kappa_i := \frac{H(\mathcal{H})}{s_i L_*} \le C_{f,D} \exp\left(-\frac{4}{5}\beta_i \log T + O_{f,D}(M^6)\right) \le e^{-M^3}. \tag{6.16}$$

Indeed $n_i \le 2T^{6\beta_i/5}$, $r_i - r_{i+1} = 2\beta_i$, and $\beta_i \log T > M^{20}$ for $i < J$. The exponent $4\beta_i/5$ is the margin left after paying for all n_i continuations. By Lemma 5.1, the joint entropy bound and this common cost imply

$$\mathbb{E}_{p,\pi}[\log p - H(Y \bmod \pi \mid \mathcal{H})] \le (e^{-cM^3} + \kappa_i)L_*. \tag{6.17}$$

Checking the transfer inequalities. In Theorem 5.2 take

$$(\tau, \delta, \tau', \beta, \beta', \eta, \eta') = (\tau_i, \delta_i, \tau_{i+1}, \beta_i, \beta_{i+1}, \eta_i, \eta_{i+1}).$$

The induction hypothesis supplies the conditional coverage at every exact value of A. Equation (6.10) supplies its joint entropy input with $\epsilon_i = e^{-cM^3}$. The parameter estimates give

$$\epsilon_i + \kappa_i \le \eta_i \delta_i \beta_{i+1}/16, \qquad \eta_{i+1} = \eta_i \delta_i/100,$$

for sufficiently large M, which are the two entropy budget requirements. For its concentration requirement, $p \le 2T$ gives

$$n_i p^{-\beta_i} \ge 2^{-\beta_i} T^{\beta_i/5} \ge \frac{1}{2} \exp(M^{20}/5).$$

It follows that $\delta_i^2 n_i p^{-\beta_i}/2 \ge \exp(M^{20}/5 - O(M))$. This quantity dominates both $\log p \le 1.05\,(100d)^M + \log 2$ and $\log(100/\delta_i) = O(M)$. Consequently

$$p \exp(-\delta_i^2 n_i p^{-\beta_i}/2) \le \delta_i/100.$$

The two remaining inequalities are

$$\delta_i^{-1} < p^{\beta_{i+1}/2}, \qquad \log(2e/\delta_i) \le (\beta_{i+1}/4) \log T.$$

They follow from $\log(1/\delta_i) = O(M)$ and $\beta_{i+1} \log T > M^{20}/2$. All hypotheses of Theorem 5.2 now hold. It proves coverage at $A_i = a$, and this start was arbitrary. Downward induction establishes the proposition uniformly over the whole schedule. □

At A_0, the conclusion gives mass at least $1/(4p)$ outside fewer than $p^{1-1/(200d)}$ residues for at least 99% of the root coordinates, from every exact start. The next section will use this coverage and the common terminal-time law to test candidate starting residues under zero avoidance. There the repeated tests will be independent conditional on residue vectors, whose laws mix exact starts; that conditioning will be treated separately.

7 Zero avoidance costs information

The sampling and coverage results hold for every self-avoiding walk with step bound D. We now assume that the walk lies in $\mathcal{A}_H(\mathcal{P}_M)$, where $\mathcal{P}_M$ is the union of the prime batches in (6.1). Thus it avoids zero modulo all root coordinates of every selected rational prime. This restriction forces short words of increments to reveal information about the starting residues. The purpose of this section is to quantify that information at the common terminal time t_*.

For each batch T, choose independently a uniform subset of q_T rational primes, with independent uniform root labels. Denote all these auxiliary choices by $\mathcal{C}$. They are independent of the walk-sampling experiment. Entropies below are always computed with $\mathcal{C}$ fixed; an expectation $\mathbb{E}_{\mathcal{C}}$ averages the resulting entropies and does not include the choices as part of the random data.

For a nonnegative integer time t, write

$$W_L(t) = (z_{t+1} - z_t, \dots, z_{t+L} - z_{t+L-1})$$

for the next L increments. Each letter has at most K_D possible values. Use the dyadic lengths

$$L_T = 2^{\lfloor \log_2(T^{1/(2d)}) \rfloor}, \qquad \tfrac{1}{2} T^{1/(2d)} < L_T \le T^{1/(2d)}.$$

These lengths are short enough that a word cannot hit one residue twice modulo a coordinate of order $p \in [T, 2T]$, yet long enough that the displacement from $A_0(T)$ has negligible entropy compared with the information in the word. Their dyadic form will let us partition longer words into shorter ones in Section 8.

Let X'_T be the selected residue vector of batch T at z_{t_*}, and let O'_T contain the selected residues of all smaller batches at that same point.

Proposition 7.1 (Information cost of one batch)**.** *There is a constant $c_1 > 0$, depending only on the fixed schedule parameters, such that for all sufficiently large M and every batch in* (6.1)*,*

$$\frac{1}{L_T}\,\mathbb{E}_{\mathcal{C}} I(X'_T; W_{L_T}(t_*) \mid O'_T) \geq \frac{c_1}{\log T}. \tag{7.1}$$

The threshold for M is uniform over all infinite self-avoiding D-step walks in $\mathcal{A}_H(\mathcal{P}_M)$.

Proof. Fix a batch, and abbreviate $L = L_T$, $q = q_T$, and $A_0 = A_0(T)$. Let X be this batch's selected residue vector at z_{A_0}, and let O contain the selected residues of all smaller batches at z_{A_0}. Their counterparts at z_{t_*} are X', O'. Put

$$s = z_{t_*} - z_{A_0}, \qquad W = W_L(t_*).$$

We will first show that repeated observations of (s, W) reveal a fixed fraction of the entropy of X left after O is known. Then we will charge the displacement entropy separately and move the result to (X', O', W).

By (6.4) and (6.5),

$$\mathbb{E}_{\mathcal{C}} H(X \mid O) \geq (1 - 2e_0) q L_*. \tag{7.2}$$

Indeed, $H(X \mid O) \geq H(X) - H(O)$, and the logarithm of the alphabet size of O is at most $e_0 c_0 T \leq e_0 q L_*$. The displacement estimate (6.9) gives

$$H(s) = O_{f,D}(T^{r_0}) = O_{f,D}(T^{1/(4d)}). \tag{7.3}$$

Here s has the original schedule law, which is independent of $\mathcal{C}$.

Testing possible checkpoint residues. Choose a sufficiently large fixed constant K_0, specified below, and set

$$m_T = \lceil K_0 (T/L) \log T \rceil .$$

For fixed $\mathcal{C}$, first draw (O, X) with its original law. Conditional on those values, draw m_T independent copies $P_a = (s_a, W^{(a)})$ of (s, W); write $P = (P_1, \ldots, P_{m_T})$. A conditional law at a value of (O, X) of probability zero may be chosen arbitrarily, since it does not affect this experiment. Each one-package marginal, together with (O, X), is the original schedule law.

For a selected root coordinate $\pi = (p, j)$, a candidate $x \in \mathbb{F}_p$ passes package a if

$$x + s_a + \sum_{h=0}^{l-1} W_h^{(a)} \not\equiv 0 \pmod{\pi} \qquad (0 \leq l < L),$$

where words are indexed from zero and the empty sum is zero. Let $\mathcal{L}_\pi(P)$ be the candidates that pass every package. The true coordinate X_π always belongs to this list: the package's exact start

has the prescribed residue X_π, and its subsequent positions lie on the zero-avoiding walk. Thus, almost surely, these lists are nonempty, and

$$H(X \mid O, P) \le \mathbb{E} \sum_{\pi \text{ selected}} \log |\mathcal{L}_\pi(P)|. \tag{7.4}$$

For each realized (O, P), the conditional support of X is contained in the Cartesian product of these lists. No independence between the coordinates of X is needed.

Coverage after conditioning on residue data. The packages just constructed are independent given (O, X), whereas the continuations used in Proposition 6.2 were sampled from an exact checkpoint time. To connect these two experiments, fix $\mathcal{C}$ and a value of (O, X) of positive probability. Its conditional package law is a mixture over the posterior exact times a at A_0. At each such exact time the future law is the original continuation law: (O, X) is determined by the time and hence imposes no further condition on the fresh continuation randomness. More explicitly, if K_a is the fresh continuation law of (s, W) from the exact time a, then for every set E of packages,

$$\Pr\{(s, W) \in E \mid O, X, \mathcal{C}\} = \sum_a \Pr\{A_0 = a \mid O, X, \mathcal{C}\}\, K_a(E). \tag{7.5}$$

The kernel K_a does not depend on the auxiliary prime choices.

Call a selected root coordinate π *favorable* for these residue data if the posterior probability that (6.15) holds at the exact A_0 start is at least $1/2$. Let $\chi(a, \pi)$ indicate failure of coverage for π at the exact start a. An unfavorable coordinate has posterior bad probability greater than $1/2$, so its indicator is at most twice that probability. The tower property gives

$$\begin{aligned} \mathbb{E}_{\mathcal{C}}\mathbb{E}_{O,X}\#\{\text{unfavorable selected coordinates}\} &\le 2\mathbb{E}_{\mathcal{C}}\mathbb{E}_{O,X} \sum_{\pi \text{ selected}} \mathbb{E}[\chi(A_0, \pi) \mid O, X, \mathcal{C}] \\ &= 2\mathbb{E}_{A_0}\mathbb{E}_{\mathcal{C}} \sum_{\pi \text{ selected}} \chi(A_0, \pi) \le 2\eta_0 q. \end{aligned} \tag{7.6}$$

The prior law of A_0 is independent of $\mathcal{C}$; its posterior law need not be. At each exact time, Proposition 6.2 bounds the fraction of bad root coordinates by η_0, and a uniform root-labelled subset of size q therefore contains at most $\eta_0 q$ bad coordinates on average. This proves the last inequality.

Fix a favorable coordinate and let μ be the posterior law of its exact start. Write G for the starts where coverage holds, so $\mu(G) \ge 1/2$. At every $a \in G$, translating the terminal residue law by $z_a \bmod \pi$ gives

$$\Pr\{s \equiv -x \pmod{\pi} \mid A_0 = a\} \ge \tau_0/p$$

for all but fewer than $p^{1-\beta_0}$ candidates x. Define

$$u_x = \mu\left\{a \in G : \Pr\{s \equiv -x \pmod{\pi} \mid A_0 = a\} \ge \tau_0/p\right\}.$$

Averaging the exceptional-set sizes yields

$$\sum_{x \in \mathbb{F}_p} (\mu(G) - u_x) < p^{1-\beta_0}.$$

If $u_x < 1/4$, the summand exceeds $1/4$. Consequently all but at most $4p^{1-\beta_0}$ candidates satisfy $u_x \ge 1/4$. We next show that each of these candidates is unlikely to survive all the packages.

A short word supplies disjoint chances of rejection. For a candidate with $u_x \geq 1/4$, take an exact start counted by u_x. The smoothing estimate (6.2), applied at that start, shows for every $0 \leq l < L$ that

$$\Pr\{z_{t_*+l} - z_{A_0} \equiv -x \pmod{\pi} \mid A_0 = a\} \geq \tau_0/(2p).$$

Indeed $l \leq T_+$, and $T_+^{-9} \leq \tau_0/(2p)$ for all batches once M is sufficiently large. Mixing over the posterior starts gives hit probability at least $\tau_0/(8p)$ at each offset.

For a single package, these L hit events are disjoint. If two offsets hit, the two walk positions have equal residues modulo π. Their difference is nonzero by self-avoidance, and hence has length at least $(p/C_f)^{1/d}$. Its length is also at most $DL < (p/C_f)^{1/d}$, since $L \leq T^{1/(2d)}$ and $p \geq T$. This contradiction establishes disjointness. One conditional package therefore rejects the candidate with probability at least cL/p, where $c = \tau_0/8$.

Independence of the packages given (O, X) now yields, for every favorable coordinate,

$$\mathbb{E}[|\mathcal{L}_\pi(P)| \mid O, X] \leq 4p^{1-\beta_0} + p\exp(-cLm_T/p) \leq 5p^{1-\beta_0}. \tag{7.7}$$

Choose the fixed K_0 large enough for the second inequality; this is possible because $T \leq p \leq 2T$ and $Lm_T/p \geq (K_0/2)\log T$. The choices and the lower threshold for M are uniform in the residue data and the walk.

By Jensen's inequality, the expected logarithm of a favorable list's size is at most $(1-\beta_0)\log p + \log 5$. For an unfavorable coordinate use $\log p$. Equations (7.4)–(7.7) imply

$$\begin{aligned}\mathbb{E}_{\mathcal{C}} H(X \mid O, P) &\leq (1-\beta_0)qL_* + 2\eta_0\beta_0 q\log(2T) + q\log 5 \\ &\leq (1-\beta_0/2)qL_*\end{aligned}$$

for large M. Here $\eta_0 = 1/100$, $L_* \geq \log T$, and $\log 5/L_* \to 0$. Subtracting this bound from (7.2) and using $e_0 = \beta_0/50$, we obtain

$$\mathbb{E}_{\mathcal{C}} I(X; P \mid O) \geq (\beta_0/3)qL_*. \tag{7.8}$$

Repeated tests have thus exposed a fixed fraction of the checkpoint residue entropy. It remains to place the cost in the terminal word itself, since only terminal words will enter the common budget.

Transporting the information to the terminal time. For fixed $\mathcal{C}$, conditional independence gives

$$\begin{aligned}I(X; P \mid O) &= H(P \mid O) - \sum_{a=1}^{m_T} H(P_a \mid O, X) \\ &\leq \sum_{a=1}^{m_T} I(X; P_a \mid O) = m_T I(X; s, W \mid O).\end{aligned}$$

There are two costs in removing the displacement. First, $I(X; s \mid O) \leq H(s)$. Second, given the full lattice displacement s, residue addition is a bijection between (O, X) and (O', X'), so

$$I(X; W \mid O, s) = I(X'; W \mid O', s).$$

For any finite variables,

$$I(A; B \mid C, S) - I(A; B \mid C) = I(A; S \mid B, C) - I(A; S \mid C) \leq H(S).$$

Removing the conditioning on s therefore costs at most another $H(s)$. In total,

$$I(X; s, W \mid O) \le I(X'; W \mid O') + 2H(s). \tag{7.9}$$

The one-package marginal is the original schedule law. Thus the variables on the right are precisely $X'_T, O'_T, W_{L_T}(t_*)$, under the same terminal law as for every other batch.

Finally, $qL_* \ge c_0 T$ and $m_T \le 2K_0(T/L)\log T$ for large T. Equations (7.8), (7.9), and (7.3) give

$$\mathbb{E}_{\mathcal{C}} I(X'; W \mid O') \ge \frac{\beta_0 c_0}{6K_0} \frac{L}{\log T} - O_{f,D}(T^{1/(4d)}).$$

Since $T^{1/(4d)} = o(L/\log T)$, this proves (7.1) with, for example, $c_1 = \beta_0 c_0/(12K_0)$. □

8 The common-law entropy budget

Each batch now charges a positive amount of information to an increment word. We must show that these charges can be added, despite the different word lengths. The residue vectors will be nested in increasing batch order. Knowing the earlier increments updates the residue vector to the start of each new block, while the smoothing of t_* compares that block's law with the law at t_*. These two facts make the entropy differences telescope.

Lemma 8.1 (A common-law entropy telescope)**.** *For every fixed collection $\mathcal{C}$ of prime choices,*

$$\sum_T \frac{I(X'_T; W_{L_T}(t_*) \mid O'_T)}{L_T} \le \log K_D + o(1), \tag{8.1}$$

where the error tends to zero uniformly over the choices and the walks as $M \to \infty$.

Proof. Order the batches as $T_1 < \cdots < T_{n_B}$, and put $L_j = L_{T_j}$. For every time t, let $Q_j(t)$ contain all selected residues from batches through T_j at z_t; let Q_0 be empty. Define

$$a_j = \frac{H(W_{L_j}(t_*) \mid Q_{j-1}(t_*))}{L_j}, \qquad b_j = \frac{H(W_{L_j}(t_*) \mid Q_j(t_*))}{L_j}.$$

The j-th charge is $a_j - b_j$. We claim that the initial entropy rate for batch $j+1$ is bounded, up to a small error, by the remaining rate b_j after batch j.

Both lengths are powers of two and $L_j \le L_{j+1}$, so L_j divides L_{j+1}. Partition the longer word into blocks of length L_j. For a block starting at offset a, $Q_j(t_*)$ and the preceding increments determine $Q_j(t_* + a)$, by adding their sum in every residue coordinate. The entropy chain rule, followed by removal of conditioning, gives

$$H(W_{L_{j+1}}(t_*) \mid Q_j(t_*)) \le \sum_{\substack{0 \le a < L_{j+1} \\ L_j \mid a}} H(W_{L_j}(t_* + a) \mid Q_j(t_* + a)).$$

Both the block and its starting residue vector are deterministic functions of the starting time on the fixed walk. By (6.2), their joint law at $t_* + a$ differs from that at t_* in total variation by at most T_+^{-9}, since $a < L_{j+1} \le T_+^{1/(2d)} \le T_+$. This is approximate invariance of the sampled time law; no stationarity of the walk is assumed.

The logarithm of the full residue alphabet is at most

$$\sum_{T} k_T \log(2T) = O(T_+),$$

by (3.2) and geometric spacing of the batches. The block alphabet adds only $L_j \log K_D = O_{f,D}(T_+)$. Lemma 3.3 therefore gives, uniformly in the block and in $\mathcal{C}$,

$$H(W_{L_j}(t_* + a) \mid Q_j(t_* + a)) \le H(W_{L_j}(t_*) \mid Q_j(t_*)) + \epsilon_M,$$
$$\epsilon_M = O_{f,D}(T_+^{-7}).$$

Indeed the continuity bound is $2h(T_+^{-9}) + O_{f,D}(T_+^{-8})$, which is smaller than the displayed error for sufficiently large M. Dividing the block sum by L_{j+1} now yields

$$a_{j+1} \le b_j + \epsilon_M / L_j \le b_j + \epsilon_M.$$

There are $n_B = O(\log T_+)$ batches, so

$$\sum_{j=1}^{n_B} (a_j - b_j) \le a_1 - b_{n_B} + (n_B - 1)\epsilon_M \le \log K_D + o(1).$$

Here $b_{n_B} \ge 0$, and the K_D-letter increment alphabet gives $a_1 \le \log K_D$. □

Proof of Theorem 1.2. Fix f, D, S and all schedule constants. Suppose that, for a given sufficiently large M, an infinite self-avoiding D-step walk lies in $\mathcal{A}_H(\mathcal{P}_M)$. Averaging Lemma 8.1 over $\mathcal{C}$ and applying Proposition 7.1 gives

$$c_1 \sum_{T} \frac{1}{\log T} \le \log K_D + o(1). \tag{8.2}$$

For each $\lceil M/2 \rceil \le m \le M$, the interval $[X_m, 1.05X_m]$ contains at least $cX_m/\log B$ of the numbers $\log T = j \log B$, for an absolute $c > 0$ and all sufficiently large M. Its contribution to the sum in (8.2) is at least $c/(1.05 \log B)$. There are order M such disjoint intervals. Their total contribution tends to infinity, whereas c_1, B, and K_D are fixed. This contradicts (8.2) once M is sufficiently large.

Every estimate and every lower threshold for M is uniform over the hypothetical avoiding walks. Choose one such M, depending only on f, D, S, and set $\mathcal{P}_D = \mathcal{P}_M$. This is one finite set of rational primes, with all root coordinates imposed in $\mathcal{A}_H(\mathcal{P}_D)$, for which no infinite avoiding walk exists. The auxiliary root-labelled subsets were used only to average the information inequalities; they do not make the final sieve depend on the walk. This proves Theorem 1.2, including its arbitrary finite exclusion set S. □

9 Completion of the classification

Theorem 9.1 (Binary-form moat classification)**.** *Let $F \in \mathbb{Z}[X, Y]$ be nonzero and homogeneous of positive degree. Then* $\mathrm{UM}(F)$ *holds if and only if F is not a nonzero integer multiple of a positive power of one nonzero integer linear form.*

Proof. Proposition 2.2 proves the necessity. For a nonexceptional F, Proposition 2.5 gives either two independent linear divisors or an irreducible homogenized divisor q^{h} of degree at least two. The first case follows from Theorem 2.4. In the second case the finite-sieve theorem, now proved, and Proposition 2.8 give $\mathrm{UM}(q^{\mathrm{h}})$. Lemma 2.1 transfers this to F. □

In particular the conclusion includes XY, $X^2 + Y^2$, $X^3 - 2Y^3$, and $X^d - 2Y^d$ for every $d \geq 2$. The latter univariate polynomials are irreducible by Eisenstein's criterion at 2. The component bound is an existence bound; no effective numerical threshold is claimed. The restriction to homogeneous forms is essential for this classification: $XY + 1$ is constantly 1 on $(n, 0)$, giving an infinite unit-step walk at prime-factor budget zero.